

Testing the Peak-Anchored Graph Pipeline

Weekly Update — 27 February 2026

Girish Narayanan

This Week's Goal

Implement and stress-test the dynamics-constrained inversion pipeline on IS-901.

Test case: 500 observations, $\omega = (0.5, -0.3, 2.0)^\circ/\text{s}$, known geometry + inertia.

Strategy: Systematic experiments — probe each component, identify failure modes, refine the approach.

Building Block Results

Question	Finding
How selective are peaks?	~0.1% of SO(3) matches brightness (80× more selective than generic epochs)
Candidate quality from sampling?	1M → ~1,400/peak, nearest ~5.95° from truth. 10M → nearest 2.14°. Scales as $N^{-1/3}$.
Bridge feasibility (arrival mismatch)?	At full scale, L-BFGS-B connects 98.5% of pairs to <0.02° mismatch. No discrimination.
Single-peak optimisation?	L-BFGS-B basins far from truth regardless of seed count. Random sampling better for coverage.

Full Pipeline Result (Steps 1–5)

50 candidates × 3 peaks. L-BFGS-B bridge optimisation. Lo-fi intermediate brightness scoring. Graph search.

Stage	Result
Candidate generation	1,400–2,987 hits/peak from 1M samples
Bridge optimisation (arrival mismatch)	98.5% feasible — 3 free ω DOF connects nearly anything
Lo-fi intermediate scoring	Truth bridge indistinguishable from random
Graph search	Truth ranked #11,979 / 121,326

The pipeline failed. But the failure pointed to something important.

Root Cause #1: Lo-fi Can't Rank

Lo-fi brightness (no shadow modelling) has attitude-dependent errors larger than the discriminating signal.

Test: Propagate from epoch 0 with true ω , truth vs 2°-nudged:

- Truth vs observed: RMS = **0.049 mag** (noise floor)
- 2°-nudged vs observed: RMS = **0.218 mag** (4.4× noise)

Hi-fi clearly sees the difference. Lo-fi cannot.

→ **Intermediate scoring must use hi-fi.** Lo-fi is fine for coarse candidate filtering, but bridge ranking requires ray-traced brightness.

Root Cause #2: The ω Degeneracy

Examined the truth bridge (nearest-to-truth candidates at peaks 1 and 2):

	True ω	Optimised ω
Value (°/s)	(0.50, -0.30, 2.00)	(0.05, 0.03, -0.32)
Magnitude	2.08 °/s	0.33 °/s
Arrival mismatch	—	0.0007° ✓
Intermediate trajectory	Correct	Completely wrong

The optimizer found a **different ω** that arrives at the right attitude (0.0007° mismatch) but takes a completely different path. It's 8× too slow and points in a different direction.

Arrival mismatch is degenerate — many ω values connect the same pair via different routes.

What This Means for the Pipeline

The original pipeline (Steps 4–5) assumes arrival mismatch + rate priors can select the right path, with brightness fitting deferred to global refinement (Step 6).

This doesn't work. By the time you reach Step 6, the graph search has already selected a wrong path based on degenerate arrival costs.

The fix: **ω must be optimised for brightness, not arrival mismatch.**

The bridge cost needs to include intermediate lightcurve fit — the only signal that distinguishes the true ω from degenerate alternatives.

Proposed Fix: Brightness-Aware Bridge Optimisation

Current (Steps 4-5):

$$\min_{\omega} \|q(t_2; q_1, \omega) - q_2\|^2$$

→ Finds any ω that arrives correctly. Degenerate.

Proposed:

$$\min_{\omega} \sum_{k \in \text{intermediate}} \|y_k - L(q(t_k; q_1, \omega))\|^2 + \lambda \|q(t_2; q_1, \omega) - q_2\|^2$$

→ ω must produce a trajectory that matches brightness **along the way**, not just at the endpoints.

Computational cost: Each ω evaluation requires Euler propagation + hi-fi brightness at ~10 intermediate epochs. At ~130ms/eval, each bridge optimisation ≈ 1.3 s. With multi-start (5 initial ω) $\times$ 5,000 bridges $\div$ 8 workers $\approx$ **70 min**. Expensive but feasible.

Hi-fi Feasibility (Confirmed)

- **Cost per hi-fi evaluation:** ~130ms (ray tracing + shadow casting)
- **Fork-based multiprocessing works:** SPICE kernels + satellite geometry survive fork
- **Scaling:** 8 workers give near-linear speedup on large batches

The bottleneck is not whether hi-fi is possible — it's where in the pipeline to spend the compute.

Summary & Next Steps

Established this week:

-  Peaks are excellent anchor points (80× selectivity)
-  Full Euler dynamics working for bridging
-  Hi-fi brightness can discriminate 2° errors (4.4× noise)
-  Hi-fi parallelisation confirmed feasible
-  Arrival mismatch is degenerate — cannot select the right ω
-  Lo-fi brightness lacks the fidelity to distinguish true from false bridges

Identified key modification to Steps 4–5:

Optimise ω for intermediate brightness fit, not arrival mismatch. This breaks the degeneracy.

Next: Implement brightness-aware bridge optimisation and rerun the graph pipeline.

AMOS 2026 — Abstract deadline: [2 March](#)